Q1 - 25 July - Shift 1

If the maximum value of the term independent of t in the expansion of $\left(t^2x^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}$, $x \geq 0$, is K, then 8K is equal to ______.

Space for your notes:

Q2 - 25 July - Shift 2

The remainder when $(11)^{1011} + (1011)^{11}$ is divided by 9 is

(A) 1 (B) 4
(C) 6 (D) 8

Space for your notes:

Q3 - 26 July - Shift 1

If the coefficients of x and x^2 in the expansion of $(1+x)^p(1-x)^q$, p, q $\leq$ 15, are -3 and -5 respectively, then the coefficient of x^3 is equal to______________.

Space for your notes:

Q4 - 26 July - Shift 2

$\sum\limits_{\substack{i,j=0 \\ i\neq j}}^{n} {}^nC_i \ {}^nC_j$ is equal to

(A) $2^{2n} - {}^{2n}C_n$ (B) $2^{2n-1} - {}^{2n-1}C_{n-1}$

(C) $2^{2n} - \frac{1}{2}\,{}^{2n}C_n$ (D) $2^{n-1} + {}^{2n-1}C_n$

Space for your notes:

Q5 - 27 July - Shift 1

The remainder when $(2021)^{2022} + (2022)^{2021}$ is divided by 7 is

(A) 0
(B) 1
(C) 2
(D) 6

Space for your notes:

Q6 - 27 July - Shift 2

Let for the 9^{th} term in the binomial expansion of $(3 + 6x)^n$, in the increasing powers of 6x, to be the greatest for $x = \frac{3}{2}$, the least value of n is n_0. If k is the ratio of the coefficient of x^6 to the coefficient of x^3, then $k + n_0$ is equal to:

Space for your notes:

Q7 - 28 July - Shift 1

The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is:

(A) 0 (B) 2 (C) 3 (D) 4

Space for your notes:

Q8 - 28 July - Shift 2

Let the coefficients of the middle terms in the expansion of $\left(\frac{1}{\sqrt{6}} + \beta x\right)^4, (1 - 3\beta x)^2$ and $\left(1 - \frac{\beta}{2}x\right)^6, \beta > 0$, respectively form the first three terms of an A.P. If d is the common difference of this A.P., then $50 - \frac{2d}{\beta^2}$ is equal to ______

Space for your notes:

Q9 - 28 July - Shift 2

If $1+(2+{}^{49}C_1+{}^{49}C_2+\ldots\ldots+{}^{49}C_{49})({}^{50}C_2+{}^{50}C_4+\ldots\ldots+{}^{50}C_{50})$ is equal to $2^n.m$, where m is odd, then n + m is equal to _____

Space for your notes:

Q10 - 29 July - Shift 1

Let the ratio of the fifth term from the beginning to the fifth term from the end in the binomial expansion of $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^n$, in the increasing powers of $\frac{1}{\sqrt[4]{3}}$ be $\sqrt[4]{6}:1$. If the sixth term from the beginning is $\frac{\alpha}{\sqrt[4]{3}}$, then α is equal to _______.

Space for your notes:

Q11 - 29 July - Shift 2

If $\sum_{k=1}^{10} K^2\left(10_{C_K}\right)^2=22000L$, then L is equal to ___.

Space for your notes:

Answer Key

Q1 (6006)	**Q2** (D)	**Q3** (23)	**Q4** (A)
Q5 (A)	**Q6** (24)	**Q7** (C)	**Q8** (57)
Q9 (99)	**Q10** (84)	**Q11** (221)	

Q1 (6006)

$$\left(t^2x^{\frac{1}{5}}+\frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}$$

$$T_{r+1}={}^{15}C_r\left(t^2x^{\frac{1}{5}}\right)^{15-r}\cdot\frac{(1-x)^{\frac{r}{10}}}{t^r}$$

For independent of t,

$30-2r-r=0$

$\Rightarrow r=10$

So, Maximum value of ${}^{15}C_{10}\ x(1-x)$ will be at

$x=\frac{1}{2}$

i.e. 6006

Q2 (D)

(D)

Q3 (23)

Since coefficient of x is -3

$\Rightarrow {}^pC_1 - {}^qC_1 = -3$

$\Rightarrow p - q = -3 \quad ....(1)$

Comparing coefficients of x^2

$-{}^pC_1{}^qC_1 + {}^pC_2 + {}^qC_2 = -5$

$$-pq + \frac{p(p-1)}{2} + \frac{q(q-1)}{2} = -5 \quad ....(2)$$

Solving (1) and (2)

$p = 8,\ q = 11$

Coefficient of x^3 is

$-{}^qC_3 + {}^pC_3 + {}^pC_1{}^qC_2 - {}^pC_2{}^qC_1$

$= -{}^{11}C_3 + {}^8C_3 + {}^8C_1{}^{11}C_2 - {}^8C_2{}^{11}C_1$

$= 23$

Q4 (A)

$$\sum_{\substack{i,j=0 \\ i\neq j}}^{n} {}^nC_i\ {}^nC_j$$

$$= \sum_{i=0}^{n} {}^nC_i \cdot \sum_{j=0}^{n} {}^nC_j - \sum_{i=j=0}^{n} \left({}^nC_i\right)^2$$

$$= (2^n)(2^n) - {}^{2n}C_n$$

$$= 2^{2n} - {}^{2n}C_n$$

Q5 (A)

$(2021)^{2022} + (2022)^{2021}$

$= (2023 - 2)^{2022} + (2023 - 1)^{2021}$

$= 7n_1 + 2^{2022} + 7n_2 - 1$

$= 7(n_1 + n_2) + 8^{674} - 1$

$= 7(n_1 + n_2) + (7 - 1)^{674} - 1$

$= 7(n_1 + n_2) + 7n_3 + 1 - 1$

$= 7(n_1 + n_2 + n_3)$

$\therefore$ Given number is divisible by 7 hence remainder is zero

Q6 (24)

$(3 + 6x)^n = {}^nC_0 3^n + {}^nC_1 3^{n-1}(6x)^1 + \dots$

$T_{r+1}\ {}^nC_r 3^{n-r}.(6x)r = {}^nC_r 3^{n-r}.6^r.\ x^r$

$= {}^nC_r 3^{n-r}.3^r.2^r.\left(\frac{3}{2}\right)^r = {}^nC_r 3^n.3^r \qquad [\text{for } x = \frac{3}{2}]$

T_9 is greatest of $x = \frac{3}{2}$

So, $T_9 > T_{10}$ and $T_9 > T_8$

(concept of numerically greatest term)

Here, $\frac{T_9}{T_{10}} > 1$ and $\frac{T_9}{T_8} > 1$

$\Rightarrow \frac{{}^nC_8 3^n.3^8}{{}^nC_9 3^n.3^9} > 1$ and $\frac{{}^nC_8 3^n.3^8}{{}^nC_7 3^n.3^7} > 1$

and $\frac{{}^nC_8}{{}^nC_7} > \frac{1}{3}$

and $\frac{n-7}{8} > \frac{1}{3}$

$\Rightarrow \frac{29}{3} < n < 11 \Rightarrow n = 10 = n_0$

So, in $(3 + 6x)^n$ for $n = n_0 = 10$

i.e., in $(3 + 6x)^{10}$, here $T_{r+1} = {}^{10}C_r 3^{10-r} 6^r x^r$

$T_7 = {}^{10}C_6 3^4.6^6.x^6 = 210.3^{10}.2^6 x^6$

$T_4 = {}^{10}C_3 3^7 6^3 x^3 = 120.3^{10}.2^3 x^3$

Ratio of coefficient of x^6 and coefficient of $x^3 = k$

$\therefore k = \frac{210.3.^{10}2^6}{120.3^{10}.2^3} = \frac{7}{4} \times 2^3 = 14$

So, $k + n_0 = 14 + 10 = 24$

Q7 (C)

$$7^{2022}+3^{2022}$$

$$=(49)^{1011}+(9)^{1011}$$

$$=(50-1)^{1011}+(10-1)^{1011}$$

$$=5\lambda-1+5K-1$$

$$=5m-2$$

$$\text{Remainder }=5-2=3$$

Q8 (57)

$${}^{4}C_{2}\times\frac{\beta^{2}}{6},-6\beta,-{}^{6}C_{3}\times\frac{\beta^{3}}{8}\text{ are in A.P}$$

$$\beta^{2}-\frac{5}{2}\beta^{3}=-12\beta$$

$$\beta=\frac{12}{5}\text{ or }\beta=-2\ \therefore\beta=\frac{12}{5}$$

$$d=-\frac{72}{5}-\frac{144}{25}=-\frac{504}{25}$$

$$\therefore 50-\frac{2d}{\beta^{2}}=57$$

Q9 (99)

$$1+\left(1+2^{49}\right)\left(2^{49}-1\right)=2^{98}$$

$$\mathrm{m}=1,\ \mathrm{n}=98$$

$$\mathrm{m}+\mathrm{n}=99$$

Q10 (84)

$$\frac{T_5}{T_{n-3}}=\frac{{}^nC_4(2^{1/4})^{n-4}(3^{-1/4})^4}{{}^nC_{n-4}(2^{1/4})^4(3^{-1/4})^{n-4}}=\frac{\sqrt[4]{6}}{1}$$

$$\Rightarrow 2^{\frac{n-8}{4}}3^{\frac{n-8}{4}}=6^{1/4}$$

$$\Rightarrow 6^{n-8}=6$$

$$\Rightarrow n-8=1\Rightarrow n=9$$

$$T_6={}^9C_5\,(2^{1/4})^4(3^{-1/4})^5=\frac{84}{\sqrt[4]{3}}$$

$$\therefore \alpha=84$$

Q11 (221)

$$\sum_{K=1}^{10} K^2\left({}^{10}C_K\right)^2$$

$$\sum_{K=1}^{10}\left(K.{}^{10}C_K\right)^2=\sum_{K=1}^{10}\left(10.{}^{9}C_{K-1}\right)^2$$

$$=100\sum_{K=1}^{10}{}^{9}C_{K-1}.{}^{9}C_{10-K}$$

$$=100\left({}^{18}C_9\right)=100\left(\frac{18!}{9!9!}\right)$$

$$\Rightarrow 4862000=22000L$$

Hence $L = 221$